



Path following of the autonomous airship with compensation of unknown wind and modeling uncertainties

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ARTICLE INFO

Article history:

Received 25 January 2019

Received in revised form 19 May 2019

Accepted 16 August 2019

Available online 22 August 2019

Keywords:

Path following

LOS guidance law

Backstepping control

Autonomous airship

Disturbance observer

ABSTRACT

The motion control of the airship has been an attractive topic in recent years. This paper investigates the problem of the path following of the autonomous airship. Due to the characteristics of the large volume, slow speed, and low-maneuverability, the airship is sensitive to the wind. Based on the design idea of the line-of-sight (LOS) guidance law, a parameter-observer-based LOS (POBLOS) guidance law is proposed. The unknown and time-varying influence from the wind is estimated by the parameter observer and then compensated in the design of the guidance loop. In order to overcome the disturbances of the modeling uncertainties and drift forces induced by the wind, the method of the command filtered backstepping control (CFBC) combined with the disturbance observer is employed in the control loop. The disturbance observer is constructed based on the formulation of the nonlinear disturbance observer (NDO) and improved by adding the status estimation error feedback, which results in a more robust performance when modeling uncertainties and time-varying disturbances exist. The stability analysis shows that the path following error is ultimately bounded. The results of the simulation indicate that the path following system has a good performance and is robust against the influence of the wind and modeling uncertainties.

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1. Introduction

As a light-than-air (LTA) vehicle, the autonomous airship has stepped into people's horizon due to its low energy consumption and the abilities to provide surveillance and communication utilities for both military and commercial use [1]. Compared to other unmanned aerial vehicles such as airplanes and helicopters, airships require much less power because the gravity is balanced by the buoyancy [2]. In addition, solar power can be converted to a useable form by means of the solar array on the airship, which provides the required power for the equipment on board [3].

The path following of the airship is an active research topic [4–6]. The airship is required to follow the reference path specified by the waypoints at a certain speed. Due to the characteristics of the large volume, slow speed and low-maneuverability, the motion of the airship is sensitive to the wind [7]. Besides, modeling uncertainties and input constraints need to be taken into account when it comes to realistic flight [8,9]. As a result, the autonomous airship must be robust against the wind and modeling uncertainties in order to accomplish the path following mission.

Many researchers have been devoted to studying path following algorithms [4,5,10–15]. Among these algorithms, the line-of-sight (LOS) guidance scheme is efficient to achieve convergence to the reference path [11]. The LOS guidance law was applied to the straight-line path following problem of the underactuated marine craft by Fossen [12]. In order to overcome the influence of the environmental disturbances, Fossen, and Pettersen modified the LOS guidance law by adding a nonlinear integral term based on adaptive control theory [11]. The observer-based LOS guidance law is another way to deal with the disturbances. Extended state observer (ESO) was introduced into the LOS guidance law by Liu [13], where a reduced-order ESO is employed to identify the influence of disturbances. In order to realize the path following of the marine vehicle in the presence of the unknown current, Belletier combined the guidance law with an ocean current observer which provides the estimate of the ocean current [14]. Nonlinear disturbance observer and finite-time anti-windup compensators were combined in [6], and the finite-time error constrained LOS (FT-ECLOS) was applied to the path following control of the stratospheric airship.

Various studies have been carried out to address the motion control of the airship [4,9,16–19]. Azinheira designed the backstepping controller for the airship path following problem and deduced a backstepping design formulation [17]. A time-specified nonsin-

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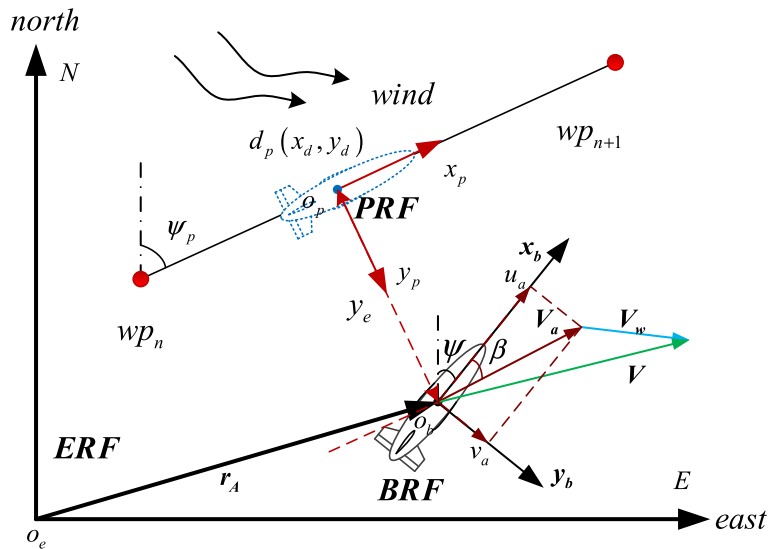


Fig. 1. The reference frames and the geometric relationship of path following problem.

gular terminal sliding mode controller was proposed by Yang to realize the trajectory tracking of the airship [9]. Zheng derived the tracking control law using the method of backstepping control [4]. In order to avoid computing command derivatives, the command filtered backstepping control (CFBC) method proposed by Farrell introduces the command filters to generate virtual signals and their first derivatives while providing magnitude, bandwidth and rate limit constraints [20]. It should be noted that the accurate model of the airship is required when CFBC is adopted, which is usually impossible in the realistic flight. The use of the disturbance observer is an efficient way to estimate and compensate the uncertainties and disturbances [21–25]. Wu combined NDO with the backstepping technique and designed the trajectory tracking controller [24]. A novel sliding mode observer (SMO) was proposed recently to estimate the disturbances, the actuator and sensor faults [25]. Kayacan proposed a self-learning disturbance observer (SLDO) used to deal with the mismatched time-varying uncertainties [23], which is based on a framework of feedback-error learning scheme [26]. Wang developed the finite-time sideslip observer (FSO) and the finite-time disturbance observer (FDO) to improve the guidance and control scheme [27].

Motivated by the previous works, the path following of the autonomous airship in the presence of the wind and modeling uncertainties is investigated in this paper. In the guidance loop, a parameter-observer-based LOS (POBLOS) guidance law is proposed. The influence of wind is expressed in terms of an unknown parameter and is estimated by the parameter observer. In the control loop, the method of CFBC is combined with the disturbance observer in order to compensate for the influence induced by the external disturbances and modeling uncertainties. It should be noted that the observers are constructed based on the formulation of NDO and modified by adding the status estimation error feedback loop which results in a more robust performance. The stability analyses of the control loop and the guidance loop are conducted. The effectiveness of the path following system is demonstrated by the simulation.

The contributions of this paper are as follows: a) Based on the design idea of NDO, the observer is modified by adding the status estimation error feedback loop which increases the robustness when the time-varying disturbances and modeling uncertainties exist during the flight. b) In the guidance loop, the influence of the wind is taken as an unknown and time-varying parameter, which is compensated by the POBLOS guidance law proposed in

this paper. c) In order to be robust to the modeling uncertainties and the drift forces induced by wind, the controllers are designed based on CFBC combined with the modified nonlinear disturbance observer (MNDO). The rest of this paper is organized as follows: Section 2 discusses the model of the autonomous airship and formulates the path following problem. In section 3, the design of the path following system is presented. Section 4 gives the stability analysis of the guidance and control system. The results of the simulation are given and the performance of the path following system is evaluated in section 5. In the last section, a brief summary is concluded.

2. Modeling and formulation

2.1. The mathematical model of the autonomous airship

The mathematical model of the autonomous airship is established according to [1]. The path following of the autonomous airship in the horizontal plane is discussed in this paper. The reference path is specified by the waypoints which are connected using straight lines. Some assumptions are taken into account in order to simplify the model of the airship [4,9]:

- The airship is under the neutral buoyancy condition

The gravity is assumed to be equal to buoyancy. In this way, the effects of gravity and buoyancy on the dynamics of the airship can be balanced. The combination of gravity and buoyancy creates a natural stabilizing moment about the roll axis as the center of buoyancy is usually above the center of gravity. In this way, the roll motion of the airship can be neglected.

- The airship is under the trimmed flight condition

The airship is assumed to be trimmed and the altitude is controlled independently. In this way, the state variables of pitch motion are small enough and their effects on the horizontal motion of the airship can be neglected.

As a result, a simplified model of the airship is obtained, which is represented by the kinematic equations and the dynamic equations [4,9,19]. The coordinate systems used in this paper are given in Fig. 1. The earth reference frame (ERF) is o_eNE , with the origin fixed on the surface of the earth: the o_eN axis and o_eE axis

point to the north and east respectively. ERF is used to describe the coordinates of the airship and the waypoints. The body reference frame (BRF) is $o_b x_b y_b$, with the origin at the center of the airship's volume; the $o_b x_b$ axis points to the head of the airship and the $o_b y_b$ axis is perpendicular to the $o_b x_b$ axis pointing to the right. The path reference frame (PRF) is used to describe the path following error. The origin of PRF is attached to the projection of the airship's position on the reference path that is determined by the waypoints wp_n and wp_{n+1} . The $o_p x_p$ axis is along with the current straight path segment. The $o_p y_p$ axis is perpendicular to the $o_p x_p$ axis and points to the right.

(1) Kinematic equation

$$\begin{cases} \dot{x} = u_a \cos \psi - v_a \sin \psi + w_n \\ \dot{y} = u_a \sin \psi + v_a \cos \psi + w_e \\ \dot{\psi} = r \end{cases} \quad (1)$$

where $\mathbf{r}_A = (x, y)^T$ denotes the coordinates of the airship in ERF and ψ is the heading angle. w_n and w_e are the components of the wind speed $\mathbf{V}_w$ in the direction of the north and east respectively. $\mathbf{V} = [u, v]^T$ represents the velocity of the airship expressed in BRF. u_a and v_a represent the components of the airspeed $\mathbf{V}_a = \mathbf{V} - \mathbf{V}_w$ expressed in BRF. The yaw rate is represented by r .

(2) Dynamic equation

Based on the assumptions mentioned above, the simplified dynamic equation is given as follows [4,19]:

$$\mathbf{M} \begin{bmatrix} \dot{u} \\ \dot{v} \\ \dot{r} \end{bmatrix} + \mathbf{C} \begin{bmatrix} u \\ v \\ r \end{bmatrix} = \boldsymbol{\tau}_a + \boldsymbol{\tau}_T + \boldsymbol{\tau}_\delta + \boldsymbol{\tau}_w \quad (2)$$

where $\mathbf{M}$ is the inertia matrix and $\mathbf{C}$ is the centrifugal and Coriolis matrix. $\boldsymbol{\tau}_a$ is the vector of the aerodynamic forces and moment. $\boldsymbol{\tau}_T$ denotes the thrust vector. $\boldsymbol{\tau}_\delta$ is the vector of the rudder control force and moment. $\boldsymbol{\tau}_w$ represents the wind-induced force vector [7]. The specific forms are given as

$$\mathbf{M} = \begin{bmatrix} m_u & 0 & 0 \\ 0 & m_v & 0 \\ 0 & 0 & I_r \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 0 & 0 & -m_v v \\ 0 & 0 & m_u u \\ m_v v & -m_u u & 0 \end{bmatrix}$$

$$\boldsymbol{\tau}_a = \begin{bmatrix} -Q \nabla^{2/3} C_x \\ -Q \nabla^{2/3} C_y + \rho V_a \nabla^{2/3} L C_{y_r}^r \\ Q \nabla^{2/3} L C_n - \rho V_a \nabla^{4/3} L^2 C_{n_r}^r \end{bmatrix}, \quad \boldsymbol{\tau}_T = \begin{bmatrix} 2F_T \\ 0 \\ 0 \end{bmatrix},$$

$$\boldsymbol{\tau}_\delta = \begin{bmatrix} 0 \\ Q \nabla^{2/3} C_{y_r}^{\delta_r} \\ -Q \nabla^{2/3} L C_{n_r}^{\delta_r} \end{bmatrix}, \quad \boldsymbol{\tau}_w = \begin{bmatrix} \mathbf{M}_{Ba} \dot{\mathbf{V}}_w + r \mathbf{M}_{Ba} \mathbf{V}_w \\ 0 \end{bmatrix}$$

where $m_u \triangleq m + \rho \nabla k_1$, $m_v \triangleq m + \rho \nabla k_2$, $I_r \triangleq I_z + \rho \nabla k_3$. m is the mass of the airship; I_z is the moment inertia; ρ stands for the atmosphere density; ∇ represents the volume of the airship; L is the length of the airship; k_1 , k_2 and k_3 are the inertia factors to calculate the added mass and inertia [1,2]. (C_x , C_y , $C_{y_r}^r$, C_n , $C_{n_r}^r$, $C_{y_r}^{\delta_r}$, $C_{n_r}^{\delta_r}$) represent the aerodynamic coefficients which are used to calculate the aerodynamic forces and moments [1,2]. V_a is the magnitude of the airspeed $\mathbf{V}_a$ and $Q = \rho V_a^2 / 2$ is the dynamic pressure of the airflow; F_T is the force produced by the propeller and δ_r is the rudder deflection. The airspeed V_a and the sideslip angle β can be obtained as

$$\begin{aligned} V_a &= \sqrt{u_a^2 + v_a^2} \\ \beta &= \sin^{-1} \frac{v_a}{V_a} \end{aligned} \quad (3)$$

$\mathbf{M}_{Ba}$ is the matrix related to the added mass and can be obtained by [7]:

$$\mathbf{M}_{Ba} = \begin{bmatrix} \rho \nabla k_1 & 0 \\ 0 & \rho \nabla k_2 \end{bmatrix} \quad (4)$$

2.2. Path following problem formulation

As illustrated in Fig. 1, the transformation matrix from ERF to PRF and can be written as

$$\mathbf{C}_e^p = \begin{bmatrix} \cos \psi_p & \sin \psi_p \\ -\sin \psi_p & \cos \psi_p \end{bmatrix} \quad (5)$$

where ψ_p is the direction of the straight line path with respect to the north and can be specified by the coordinates of waypoint $wp_n(x_n, y_n)$ and $wp_{n+1}(x_{n+1}, y_{n+1})$ in ERF as

$$\psi_p = \tan^{-1} \frac{y_{n+1} - y_n}{x_{n+1} - x_n} \quad (6)$$

Since the origin of PRF is the projection of the airship's position on the reference path, the path following error can be represented by the cross-error y_e in Fig. 1, which is used to describe the deviation from the reference path. y_e can be calculated according to the position of the airship (x, y) and the desired position $d_p(x_d, y_d)$ as follows,

$$\begin{pmatrix} 0 \\ y_e \end{pmatrix} = \mathbf{C}_e^p \begin{pmatrix} x - x_d \\ y - y_d \end{pmatrix} = \begin{pmatrix} \cos \psi_p & \sin \psi_p \\ -\sin \psi_p & \cos \psi_p \end{pmatrix} \begin{pmatrix} x - x_d \\ y - y_d \end{pmatrix} \quad (7)$$

Expanding (7) gives (8) and (9),

$$(x - x_d) \cos \psi_p + (y - y_d) \sin \psi_p = 0 \quad (8)$$

$$y_e = -(x - x_d) \sin \psi_p + (y - y_d) \cos \psi_p \quad (9)$$

The coordinates of the desired position can be calculated using the geometric relationships with waypoint $wp_n(x_n, y_n)$ and $wp_{n+1}(x_{n+1}, y_{n+1})$ in Fig. 1 and are derived briefly as follows:

It is known that the point d_p is along the straight path specified by the waypoint wp_n and wp_{n+1} . In this way, (x_d, y_d) can be represented by (x_n, y_n) and (x_{n+1}, y_{n+1}) using a scalar λ as (10)

$$\begin{cases} x_d = x_n + \lambda \cdot \Delta x \\ y_d = y_n + \lambda \cdot \Delta y \end{cases} \quad (10)$$

where $\Delta x = x_{n+1} - x_n$, $\Delta y = y_{n+1} - y_n$. According to the definition of the path following error, the error vector $\mathbf{e} = [x - x_d, y - y_d]^T$ is perpendicular to the vector $\mathbf{p} = [\Delta x, \Delta y]^T$, which can be described as

$$\Delta x(x - x_d) + \Delta y(y - y_d) = 0 \quad (11)$$

Using (10) and (11), the scalar λ can be calculated as (12)

$$\lambda = \frac{(x - x_n) \cdot \Delta x + (y - y_n) \cdot \Delta y}{\Delta x^2 + \Delta y^2} \quad (12)$$

Then, (x_d, y_d) can be obtained using (12) and (10). The path following problem of the autonomous airship can be formulated as:

Exposed to the environmental disturbances, the autonomous airship will approach to the reference path and the path following error will converge to a bounded residual set.

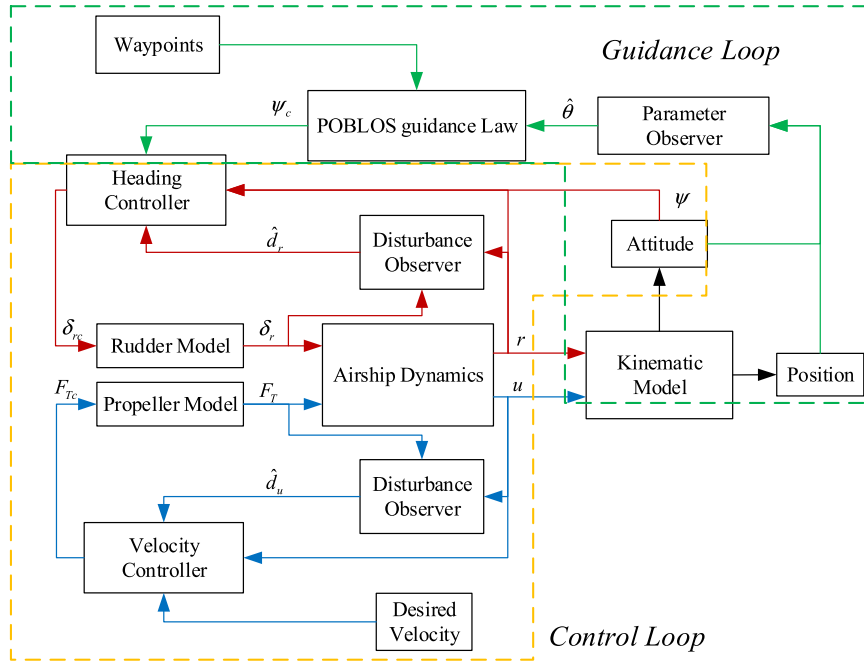


Fig. 2. The architecture of the path following system of the autonomous airship.

3. Path following system design

The path following system consists of the guidance loop and the control loop. In the guidance loop, the path following error is transformed to the heading command by POBLOS guidance law where a parameter observer is designed to estimate the influence of the wind. The control loop is made up of the heading control loop and the velocity control loop. The lumped disturbances including the external disturbances and modeling uncertainties are estimated by the disturbance observers. The heading control loop is designed to execute the heading command. The design of the velocity control loop is to guarantee that the autonomous airship follows the path in the desired velocity. The overall architecture of the path following system is shown in Fig. 2.

3.1. Design of the observer with the status estimation error feedback

During the flight of the airship, disturbances and uncertainties exist inevitably and have a significant impact on the effectiveness of the guidance and control system. The observer is usually adopted to estimate the disturbances and uncertainties in order to increase the robustness of the system.

Nonlinear disturbance observer (NDO) is usually combined with the controller design to compensate the disturbances [21,24]. In this paper, NDO is improved by adding the status estimation error feedback in order to be more robust when the time-varying disturbances and uncertainties exist during the flight.

The dynamic system is described as

$$\dot{x} = f(x) + g(x)u + d \quad (13)$$

where u and d denote the input variable and the unknown variable, respectively. x is the status variable of the dynamic system. The unknown variable d can represent the influence of the disturbances and modeling uncertainties on the motion of the airship.

Assumption 1. The unknown variable d is time-varying but its first derivative $\dot{d}$ is bounded and satisfies $|\dot{d}| < |\dot{d}|_{\max}$ where $|\dot{d}|_{\max}$ is a positive constant.

3.1.1. A brief introduction of NDO

According to the formulation of NDO in [21], NDO is constructed as

$$\begin{cases} \dot{z} = -lz - l[p(x) + f(x) + g(x)u] \\ \hat{d} = z + p(x) \end{cases} \quad (14)$$

where $\hat{d}$ and z represent the estimation of d and the internal state variable. $p(x) = l(x - x_0)$ is the designed function and x_0 is the initial value of x . l is the gain of the observer and is usually selected as a constant value.

$\tilde{d} = d - \hat{d}$ denotes the estimation error of the unknown variable d . By combining (14) and (13), the estimation error dynamic equation of NDO is given as follows

$$\dot{\tilde{d}} = \dot{d} - l\tilde{d} \quad (15)$$

By Laplace transformation of (15), we can obtain

$$\tilde{d}(s) = \frac{1}{s+l} \dot{d}(s) \quad (16)$$

where s is the Laplace operator. From (16), it can be seen that the estimation error $\tilde{d}$ of NDO is dependent on $\dot{d}$. When the unknown variable d is assumed to be constant, the estimation value $\hat{d}$ will converge to d asymptotically as illustrated in [24]. However, when the unknown variable d is time-varying, the estimation error $\tilde{d}$ exists and is bounded. The bound of the estimation error can be described as (17)

$$|\tilde{d}| < \frac{|\dot{d}|_{\max}}{l} \quad (17)$$

It should be noted that the only way to reduce the bound of $\tilde{d}$ is to increase the gain l of the observer. However, the gain l is constrained by the bandwidth of the physical system [24] and too large gain l may result in the oscillation and even instability of the system.

3.1.2. The improvement of NDO with the status estimation error feedback

NDO is improved by adding the status estimation error feedback in this paper in order to increase the robustness of the observer when the unknown variable d is time-varying. The modified nonlinear disturbance observer (MNDO) is constructed as (18)

$$\begin{cases} \dot{\hat{x}} = f(x) + g(x)u + \hat{d} + k\tilde{x} \\ \dot{\hat{z}} = -lz - l(\hat{p}(x) + f(x) + g(x)u) \\ \hat{d} = z + p(x) \end{cases} \quad (18)$$

where $\hat{x}$ and $\tilde{x} = x - \hat{x}$ denote the estimation and the estimation error of the status variable x . $\hat{p}(x) = l(\hat{x} - x_0)$ represents the estimation of the designed function $p(x) = l(x - x_0)$. k and l are the design parameters. Using (13) and (18), the estimation error dynamic equations of MNDO are as follows

$$\begin{cases} \dot{\tilde{x}} = \tilde{d} - k\tilde{x} \\ \dot{\tilde{d}} = \dot{d} - l\tilde{d} - l^2\tilde{x} \end{cases} \quad (19)$$

Considering (15) and (19), the structure diagram of the estimation error is depicted as Fig. 3.

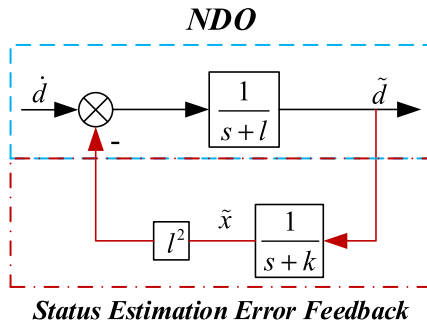


Fig. 3. The estimation error structure diagram of MNDO.

By Laplace transformation of (19), the relationship between $\tilde{d}$ and $\tilde{x}$ is described as

$$\tilde{d}(s) = \frac{s+k}{s^2 + (k+l)s + kl + l^2} \dot{d}(s) \quad (20)$$

The bound of the estimation error is specified as

$$|\tilde{d}| < \frac{k}{kl+l^2} |\dot{d}|_{\max} \quad (21)$$

Compared to NDO, MNDO increases a negative feedback loop using the estimation error of the status x . From (21), the estimation error can be reduced by selecting the parameter k and l appropriately under the constraints of the bandwidths and the dynamic performance of the observer.

Instead of the open loop structure, the status estimation error feedback loop improves the performance of NDO when the unknown variable d is time-varying.

3.1.3. The stability analysis of the observer

The Lyapunov function candidate is proposed as

$$V_d = \frac{1}{2}\tilde{x}^2 + \frac{1}{2l^2}\tilde{d}^2 \quad (22)$$

Using (19), the time differentiation of (22) yields

$$\begin{aligned} \dot{V}_d &= -k\tilde{x}^2 - \frac{1}{l}\tilde{d}^2 + \frac{1}{l^2}\tilde{d}\dot{\tilde{d}} \\ &\leq -k\tilde{x}^2 - \frac{1}{l^2}\left(l - \frac{1}{2}\right)\tilde{d}^2 + \frac{1}{2l^2}|\dot{d}|^2 \\ &\leq -2cV_d + \frac{1}{2l^2}|\dot{d}|_{\max}^2 \end{aligned} \quad (23)$$

where $c = \min(k, l - 0.5)$. The modified observer gains should satisfy: $k > 0$, $l > 0.5$.

According to Lasalle-Yoshizawa Theorem [28], the estimation errors of the status variable x and the unknown variable d are uniformly ultimately bounded. The observer is stable in the following compact

$$\Omega = \left\{ \tilde{x}, \tilde{d} \mid V_d < \frac{1}{4cl^2} |\dot{d}|_{\max} \right\} \quad (24)$$

3.2. Design of the guidance loop based on POBLOS guidance law

POBLOS guidance law is proposed by combining the parameter observer with the integral LOS guidance law [11].

Substituting (1) and (8) into the derivative of (9) gives

$$\dot{y}_e = u_a \sin(\psi - \psi_p) + u_a \cos(\psi - \psi_p)\theta \quad (25)$$

where

$$\theta = \tan \beta - \frac{w_n}{u_a} \cdot \frac{\sin \psi_p}{\cos(\psi - \psi_p)} + \frac{w_e}{u_a} \cdot \frac{\cos \psi_p}{\cos(\psi - \psi_p)} \quad (26)$$

w_n and w_e represent the components of the wind speed in the direction of the north and the east. θ denotes the influence of the sideslip angle β and the external wind. As the external wind and the sideslip angle β are usually unknown but bounded, it is reasonable to make the following assumption in order to simplify the analysis.

Assumption 2. The unknown term θ and its time derivation $\dot{\theta}$ are bounded, which indicates that there exist positive constants M_θ and $M_{\dot{\theta}}$ such that

$$\begin{cases} |\theta| < M_\theta \\ |\dot{\theta}| < M_{\dot{\theta}} \end{cases} \quad (27)$$

In this paper, a parameter observer is designed to estimate θ . The parameter observer is constructed based on the formulation given in section 3.1.2 as

$$\begin{cases} \dot{\hat{y}}_e = u_a \sin(\psi - \psi_p) + u_a \cos(\psi - \psi_p)\hat{\theta} + k_{y_e}\tilde{y}_e \\ \dot{\hat{z}}_\theta = -l_\theta u_a [\sin(\psi - \psi_p) + \cos(\psi - \psi_p)(z_\theta + l_\theta(\hat{y}_e - y_{e0}))] \\ \hat{\theta} = z_\theta + l_\theta(y_e - y_{e0}) \end{cases} \quad (28)$$

where $\hat{y}_e$ and $\hat{\theta}$ are the estimations of y_e and θ respectively. z_θ is the internal variable of the parameter observer. $\tilde{y}_e = y_e - \hat{y}_e$ and $\tilde{\theta} = \theta - \hat{\theta}$ denote the estimation errors of y_e and θ . y_{e0} is the initial value of y_e . k_{y_e} and l_θ are the design parameters.

The integral LOS guidance law [11] is adopted at a kinematic level and the goal is to obtain the heading command ψ_c as

$$\psi_c = \psi_p + \tan^{-1}\left(-\frac{y_e + \alpha_y}{\Delta}\right) \quad (29)$$

where Δ is the lookahead distance which is usually dependent on the maneuvering characteristics of the airship. ψ_p stands for the direction of the straight line path with respect to the north. α_y is the virtual control input and is designed to combine with the parameter observer to deal with the influence of the wind. The virtual control input α_y is designed as

$$\alpha_y = \Delta \hat{\theta} \quad (30)$$

The heading command ψ_c is limited in magnitude: $M_{\psi-} \leq \psi_c \leq M_{\psi+}$. The bounds $M_{\psi-}$ and $M_{\psi+}$ are specified as follows

$$\begin{cases} M_{\psi-} > \psi_p - \delta_\psi \\ M_{\psi+} < \psi_p + \delta_\psi \end{cases} \quad (31)$$

where $\delta_\psi \in (0, \pi/2)$ is an adjustable parameter, which is designed in order to guarantee that the response of the airship will not be aggressive when the path following error y_e is too large.

According to (25) and (28), the error dynamic equations of the parameter observer are given as

$$\begin{cases} \dot{\tilde{y}}_e = u_a \tilde{\theta} \cos(\psi - \psi_p) - k_{y_e} \tilde{y}_e \\ \dot{\tilde{\theta}} = \dot{\theta} - l_\theta u_a \cos(\psi - \psi_p) \tilde{\theta} - l_\theta^2 u_a \cos(\psi - \psi_p) \tilde{y}_e \end{cases} \quad (32)$$

A Lyapunov function candidate V_θ is introduced as

$$V_\theta = \frac{1}{2} \tilde{y}_e^2 + \frac{1}{2l_\theta^2} \tilde{\theta}^2 \quad (33)$$

Differentiating (33) with respect to time yields

$$\begin{aligned} \dot{V}_\theta &= -k_{y_e} \tilde{y}_e^2 + \frac{1}{l_\theta^2} \tilde{\theta} \dot{\tilde{\theta}} - \frac{1}{l_\theta} u_a \tilde{\theta}^2 \cos(\psi - \psi_p) \\ &\leq -k_{y_e} \tilde{y}_e^2 - \frac{1}{l_\theta} u_a \tilde{\theta}^2 \cos(\psi - \psi_p) + \frac{1}{l_\theta^2} \tilde{\theta} M_\theta \\ &\leq -k_{y_e} \tilde{y}_e^2 - \frac{1}{l_\theta^2} \left(u_a l_\theta \cos(\psi - \psi_p) - \frac{1}{2} \right) \tilde{\theta}^2 + \frac{1}{2l_\theta^2} M_\theta^2 \end{aligned} \quad (34)$$

Here we assume that the heading angle will track the heading command with a bounded tracking error by the heading controller in the following section and the proof will be given in section 4. As a result, the following inequality is satisfied according to (31)

$$-\delta_\psi < \psi - \psi_p < \delta_\psi \quad (35)$$

In this way, (35) will be simplified as

$$\begin{aligned} \dot{V}_\theta &\leq -k_{y_e} \tilde{y}_e^2 - \frac{1}{l_\theta^2} \left(u_a l_\theta \cos \delta_\psi - \frac{1}{2} \right) \tilde{\theta}^2 + \frac{1}{2l_\theta^2} M_\theta^2 \\ &\leq -2c_\theta V_\theta + \frac{1}{2l_\theta^2} M_\theta^2 \end{aligned} \quad (36)$$

where

$$c_\theta = \min \left(k_{y_e}, \frac{1}{l_\theta^2} \left(u_a l_\theta \cos \delta_\psi - \frac{1}{2} \right) \right)$$

Then the observer parameters are designed to satisfy: $k_{y_e} > 0$, $l_\theta > 1/(2u_a \cos \delta_\psi)$. Using Lasalle-Yoshizawa Theorem [28] again, the estimation errors $\tilde{y}_e$ and $\tilde{\theta}$ can be proved uniformly ultimately bounded and the bounds are restricted in the following compact

$$\Omega_\theta = \left\{ \tilde{y}_e, \tilde{\theta} \mid V_\theta < \frac{1}{4c_\theta l_\theta^2} M_\theta^2 \right\} \quad (37)$$

3.3. Design of the control loop based on the disturbance compensation

The control loop of the path following system consists of the heading control loop and the velocity control loop. Considering the influence of the external disturbances and the modeling uncertainties, the dynamic equations of the airship are rewritten in a standard form as

$$\begin{cases} \dot{r} = f_r + g_r \delta_r + d_r \\ \dot{u}_a = f_u + g_u F_T + d_u \end{cases} \quad (38)$$

where

$$f_r = \frac{Q \nabla^{2/3} L C_n - \rho V_a \nabla^{4/3} L^2 C_n^r r}{I_z + \rho \nabla k_3},$$

$$\begin{aligned} f_u &= \frac{m + \rho \nabla k_2}{m + \rho \nabla k_1} v r - \frac{Q \nabla^{2/3} C_x}{m + \rho \nabla k_1} \\ g_r &= \frac{-Q \nabla^{2/3} L C_n^{\delta_r}}{I_z + \rho \nabla k_3}, \quad g_u = \frac{2}{m + \rho \nabla k_1} \end{aligned}$$

g_u and g_r are the coefficients of control force and moment. f_u and f_r represent the nonlinear terms related to the other forces and moments acting on the airship. d_r and d_u denote the lumped disturbances caused by the external wind and modeling uncertainties.

The control loop is designed based on the idea of command filtered backstepping control [20] in which a command filter is adopted to generate virtual signals and their first derivatives. As a result, the analytic computation of the time derivatives of the heading command and yaw rate command can be avoided. The details of the command filter design can be found in [20].

Assumption 3. The lumped disturbance d_r and d_u are time-varying and their first derivative $\dot{d}_r$ and $\dot{d}_u$ are bounded and satisfy $|\dot{d}_r| < |\dot{d}_r|_{\max}$, $|\dot{d}_u| < |\dot{d}_u|_{\max}$ where $|\dot{d}_r|_{\max}$ and $|\dot{d}_u|_{\max}$ are positive constants.

Remark 1. According to [1], the aerodynamic forces and moments are the functions of the sideslip angle β which is usually difficult to measure when it comes to the realistic flight. In this way, the design of the controller is conducted assuming there is no wind. Then the disturbances induced by the wind are estimated and compensated with the help of the disturbance observer.

3.3.1. Heading control loop design

The heading command ψ_c is given in the guidance loop. ψ_d and $\dot{\psi}_d$ are the output variables of the heading command filter when ψ_c is taken as the input signal. Define the heading angle tracking error

$$\tilde{\psi} = \psi - \psi_d \quad (39)$$

The time derivative of $\tilde{\psi}$ is given as

$$\dot{\tilde{\psi}} = r_c + \tilde{r} + (r_d - r_c) - \dot{\psi}_d \quad (40)$$

where r_c is the yaw rate command. r_d and $\dot{r}_d$ are the output signals from the command filter when r_c is the input signal. $\tilde{r} = r - r_d$ denotes the yaw rate tracking error. The yaw rate command r_c is designed as

$$r_c = -k_\psi \tilde{\psi} + \dot{\psi}_d \quad (41)$$

where k_ψ is the control parameter to be designed. Substitution of (41) into (40) gives

$$\dot{\tilde{\psi}} = -k_\psi \tilde{\psi} + \tilde{r} + \tilde{r}_c \quad (42)$$

$\tilde{r}_c = r_d - r_c$ denotes the error of yaw rate command.

Considering (38), the time derivative of the yaw rate tracking error is written as

$$\dot{\tilde{r}} = f_r + g_r \delta_{rc} + g_r \tilde{\delta}_{rc} + d_r - \dot{r}_d \quad (43)$$

$\tilde{\delta}_{rc} = \delta_r - \delta_{rc}$ represents the error of the rudder deflection. δ_{rc} and δ_r denote the rudder deflection command and the real rudder deflection respectively. Then the rudder deflection command δ_{rc} is designed as

$$\delta_{rc} = g_r^{-1} (-f_r - k_r \tilde{r} + \dot{r}_d - \tilde{\psi} - \dot{\tilde{\psi}}) \quad (44)$$

where k_r is the design parameter. $\hat{d}_r$ is the estimation of d_r and can be obtained by MNDO mentioned above.

Remark 2. When the command limits are not in effect, the command errors $\tilde{\psi}_c = \psi_d - \psi_c$ and $\tilde{r}_c = r_d - r_c$ can be made arbitrarily small by selecting the command filter parameter sufficiently. When the limits are in effect, the command errors are bounded because the input and output signals are all bounded. The details of the characteristics of the command filter will be found in Appendix A of [20].

Remark 3. The rudder operation constraint is a practical issue in which rudder deflection and rate saturation along with its dynamics are needed to be taken into consideration. In this paper, the dynamics of the rudder is modeled as a second-order system described by the natural frequency ω_n and damping ratio ξ with constraints on rudder deflection and rate. When the command δ_{rc} is too aggressive, the error of the rudder deflection $\tilde{\delta}_{rc}$ exists and is bounded. It should be noted that when the rudder operation constraints are not in effect, $\tilde{\delta}_{rc}$ will converge to zero.

3.3.2. Velocity control loop design

The surge velocity tracking error is defined as

$$\tilde{u} = u - u_d \quad (45)$$

where u_d and $\dot{u}_d$ are the output variables of the velocity command filter when u_c is the input signal. Considering (38), the time derivative of (45) is given as

$$\dot{\tilde{u}} = f_u + g_u F_{Tc} + g_u \tilde{F}_{Tc} + \dot{u}_d - \dot{u}_d \quad (46)$$

where $\tilde{F}_{Tc} = F_T - F_{Tc}$ represents the error of thrust induced by the limits of the propellers. F_{Tc} and F_T are the thrust command and the real thrust respectively. The thrust command is designed as

$$F_{Tc} = g_u^{-1}(-k_u \tilde{u} - f_u - \dot{u}_d + \dot{u}_d) \quad (47)$$

where k_u is a design parameter. $\dot{u}_d$ is the estimation of $\dot{u}_d$ and is obtained by MNDO mentioned in section 3.1.2.

Remark 4. F_T is the force produced by the propellers. In this paper, the dynamic model of the propeller is simplified as a first-order system depicted by the time constant $\tau = 1/\omega_n$ with reference to the results from [29]. When the constraints on the magnitude and bandwidth are in effect, $\tilde{F}_{Tc}$ exists and is bounded.

4. Stability analysis

Theorem. The model of the autonomous airship is described by (1) and (2). The heading command is computed by (29) and the velocity command is set as a constant value. The rudder command and the thrust command are obtained using (44) and (47). Suppose the Assumption 1–3 are satisfied. Then the following statements hold:

- i. The tracking error $\tilde{\psi}$, $\tilde{r}$ and $\tilde{u}$ in the control loop are ultimately uniformly bounded.
- ii. The path following error y_e will ultimately converge to the compact set around the origin.

Proof. i. Using (43), (44), (46) and (47), the error dynamic equations of the control loop can be summarized as

$$\begin{cases} \dot{\tilde{\psi}} = -k_\psi \tilde{\psi} + \tilde{r} + \tilde{r}_c \\ \dot{\tilde{r}} = -k_r \tilde{r} - \tilde{\psi} + \tilde{d}_r + g_r \tilde{\delta}_{rc} \\ \dot{\tilde{u}} = -k_u \tilde{u} + g_u \tilde{F}_{Tc} + \tilde{d}_u \end{cases} \quad (48)$$

where $\tilde{d}_r = d_r - \hat{d}_r$ and $\tilde{d}_u = d_u - \hat{d}_u$ represent the estimation errors of the disturbances.

A Lyapunov function candidate for the control loop is introduced as

$$V_{con} = \frac{1}{2} \tilde{\psi}^2 + \frac{1}{2} \tilde{r}^2 + \frac{1}{2} \tilde{u}^2 \quad (49)$$

Considering (48), the time derivative of (49) is written as

$$\begin{aligned} \dot{V}_{con} &= -k_\psi \tilde{\psi}^2 - k_r \tilde{r}^2 + \tilde{\psi} \tilde{r}_c + \tilde{r} \tilde{d}_r + g_r \tilde{r} \tilde{\delta}_{rc} - k_u \tilde{u}^2 \\ &\quad + g_u \tilde{F}_{Tc} \tilde{u} + \tilde{d}_u \tilde{u} \\ &\leq -k_\psi \tilde{\psi}^2 - k_r \tilde{r}^2 - k_u \tilde{u}^2 + \frac{1}{2} \tilde{\psi}^2 + (1 + g_r) \frac{1}{2} \tilde{r}^2 \\ &\quad + (1 + g_u) \frac{1}{2} \tilde{u}^2 + \frac{1}{2} \tilde{r}_c^2 + \frac{1}{2} g_r \tilde{\delta}_{rc}^2 + \frac{1}{2} g_u \tilde{F}_{Tc}^2 + \frac{1}{2} \tilde{d}_r^2 + \frac{1}{2} \tilde{d}_u^2 \\ &\leq -\left(k_\psi - \frac{1}{2}\right) \tilde{\psi}^2 - \left(k_r - \frac{1 + g_r}{2}\right) \tilde{r}^2 - \left(k_u - \frac{1 + g_u}{2}\right) \tilde{u}^2 \\ &\quad + \frac{1}{2} \tilde{r}_c^2 + \frac{1}{2} g_r \tilde{\delta}_{rc}^2 + \frac{1}{2} g_u \tilde{F}_{Tc}^2 + \frac{1}{2} \tilde{d}_r^2 + \frac{1}{2} \tilde{d}_u^2 \\ &\leq -2c_{con} V_{con} + W_{con} \end{aligned} \quad (50)$$

where

$$c_{con} = \min\left(k_\psi - \frac{1}{2}, k_r - \frac{1 + g_r}{2}, k_u - \frac{1 + g_u}{2}\right)$$

$$W_{con} = \frac{1}{2} \tilde{r}_c^2 + \frac{1}{2} g_r \tilde{\delta}_{rc}^2 + \frac{1}{2} g_u \tilde{F}_{Tc}^2 + \frac{1}{2} \tilde{d}_r^2 + \frac{1}{2} \tilde{d}_u^2$$

The control parameters k_ψ , k_r and k_u should be designed: $k_\psi > 0.5$, $k_r > 0.5(1 + g_r)$, $k_u > 0.5(1 + g_u)$. g_r and g_u are the parameters related to the airship's characteristics and are given in (38).

According to the stability analysis of MNDO in section 3.1.3, the estimation errors $\tilde{d}_r$ and $\tilde{d}_u$ are bounded and the bound can be made small by choosing appropriate observer parameters. According to Remark 2–4, $\tilde{r}_c$, $\tilde{\delta}_{rc}$ and $\tilde{F}_{Tc}$ can be regarded as small bounded quantities. In this way, the tracking errors $\tilde{\psi}$, $\tilde{r}$ and $\tilde{u}$ are proved to be uniformly ultimately bounded using Lasalle-Yoshizawa Theorem [28]. The control loop is stable in the compact as follows

$$\Omega_{con} = \left\{ \tilde{\psi}, \tilde{r}, \tilde{u} \mid V_{con} < \frac{1}{2c_{con}} W_{con} \right\} \quad (51)$$

ii. The heading control loop can be used in cascade with the guidance loop. The effectiveness of the guidance loop is affected by the heading tracking error $\tilde{\psi}$. To see it more clearly, substituting (29) and (30) into (25) yields

$$\begin{aligned} \dot{y}_e &= u_a \sin(\psi_c - \psi_p + \psi_e) + u_a \cos(\psi_c - \psi_p + \psi_e) \theta \\ &= -u_a \frac{y_e + \alpha_y}{\sqrt{\Delta^2 + (y_e + \alpha_y)^2}} \cos \psi_e \\ &\quad + u_a \frac{\Delta}{\sqrt{\Delta^2 + (y_e + \alpha_y)^2}} \sin \psi_e \\ &\quad + u_a \frac{\Delta \theta}{\sqrt{\Delta^2 + (y_e + \alpha_y)^2}} \cos \psi_e \\ &\quad + u_a \frac{(y_e + \alpha_y) \theta}{\sqrt{\Delta^2 + (y_e + \alpha_y)^2}} \sin \psi_e \\ &= -u_a \frac{y_e}{\sqrt{\Delta^2 + (y_e + \alpha_y)^2}} + u_a \frac{\Delta \tilde{\theta}}{\sqrt{\Delta^2 + (y_e + \alpha_y)^2}} \end{aligned}$$

$$+ u_a \phi(\theta, \psi_e, y_e) \psi_e \quad (52)$$

where

$$\begin{aligned} \psi_e &= \psi - \psi_c = (\psi - \psi_d) + (\psi_d - \psi_c) \\ &= \tilde{\psi} + \tilde{\psi}_c \end{aligned}$$

$\tilde{\psi}$ and $\tilde{\psi}_c$ denote the heading tracking error and the heading command error, respectively. According to the conclusions from the stability analysis of the control loop and Remark 2, ψ_e can be regarded as a small bounded quantity. θ is the estimation error of the unknown parameter θ in POBLOS guidance law, which is bounded and can be reduced by selecting the appropriate observer parameters due to the conclusion from (37).

The function $\phi(\theta, \psi_e, y_e)$ is defined as

$$\begin{aligned} \phi(\theta, \psi_e, y_e) &= \frac{1 - \cos \psi_e}{\psi_e} \frac{y_e + \alpha_y}{\sqrt{\Delta^2 + (y_e + \alpha_y)^2}} \\ &+ \frac{\sin \psi_e}{\psi_e} \frac{\Delta}{\sqrt{\Delta^2 + (y_e + \alpha_y)^2}} \\ &- \frac{1 - \cos \psi_e}{\psi_e} \frac{\Delta \theta}{\sqrt{\Delta^2 + (y_e + \alpha_y)^2}} \\ &+ \frac{\sin \psi_e}{\psi_e} \frac{(y_e + \alpha_y) \theta}{\sqrt{\Delta^2 + (y_e + \alpha_y)^2}} \end{aligned} \quad (53)$$

The function $\phi(\theta, \psi_e, y_e)$ is bounded because $|\sin(x)/x| < 1$, $|(1 - \cos(x))/x| < 0.73$ for all x and

$$\left| \frac{y_e + \alpha_y}{\sqrt{\Delta^2 + (y_e + \alpha_y)^2}} \right| < 1, \quad \left| \frac{\Delta}{\sqrt{\Delta^2 + (y_e + \alpha_y)^2}} \right| < 1$$

Considering the boundedness of θ , an upper bound of $\phi(\theta, \psi_e, y_e)$ will be obtained as

$$|\phi(\theta, \psi_e, y_e)| < 1.73(1 + M_\theta) \quad (54)$$

where M_θ denotes the bound of the unknown parameter $|\theta|$ as described in (27).

Design the Lyapunov function candidate of the path following error as follows

$$V_e = \frac{1}{2} y_e^2 \quad (55)$$

Considering (52), differentiating (55) with respect to time gives

$$\begin{aligned} \dot{V}_e &= -u_a \frac{y_e^2}{\sqrt{\Delta^2 + (y_e + \alpha_y)^2}} + u_a \frac{\Delta y_e \tilde{\theta}}{\sqrt{\Delta^2 + (y_e + \alpha_y)^2}} \\ &+ u_a \phi(\theta, \psi_e, y_e) \psi_e y_e \\ &\leq -u_a \frac{y_e^2}{\sqrt{\Delta^2 + (y_e + \alpha_y)^2}} + u_a y_e \tilde{\theta} + u_a \phi(\theta, \psi_e, y_e) \psi_e y_e \\ &\leq -u_a \frac{y_e^2}{\sqrt{\Delta^2 + (y_e + \alpha_y)^2}} + |u_a y_e| (|\tilde{\theta}| + 1.73(1 + M_\theta) \psi_e) \\ &\leq -u_a (1 - \sigma) \frac{y_e^2}{\sqrt{\Delta^2 + (y_e + \alpha_y)^2}} \\ &- |u_a y_e| \left(\frac{\sigma |y_e|}{\sqrt{\Delta^2 + (y_e + \alpha_y)^2}} - M_e \right), \quad \sigma \in (0, 1) \end{aligned} \quad (56)$$

where $M_e \triangleq |\tilde{\theta}| + 1.73(1 + M_\theta) \psi_e$. $|\tilde{\theta}|$ denotes the magnitude of the estimation error of the unknown parameter θ . $\sigma \in (0, 1)$ is a positive constant.

Considering (29) and (31), we can obtain

$$(y_e + \alpha_y)^2 < \Delta^2 \tan^2 \delta_\psi \quad (57)$$

where $\delta_\psi \in (0, \pi/2)$ is an adjustable parameter in (31) and used to constrain the magnitude of the heading command. In this way, (56) will be simplified as (58)

$$\begin{aligned} \dot{V}_e &\leq -u_a (1 - \sigma) \frac{y_e^2}{\Delta \sqrt{1 + \tan^2 \delta_\psi}} - |u_a y_e| (f(y_e) - M_e), \\ \sigma &\in (0, 1) \end{aligned} \quad (58)$$

$f(y_e)$ is an even function of the path following error y_e with $f(0) = 0$

$$f(y_e) = \frac{\sigma |y_e|}{\Delta \sqrt{1 + \tan^2 \delta_\psi}}$$

If $f(y_e) > M_e$, then

$$\dot{V}_e < -u_a (1 - \sigma) \frac{y_e^2}{\Delta \sqrt{1 + \tan^2 \delta_\psi}} \quad (59)$$

Based on the conclusion of the input-to-state stability from [30], the path following error y_e is bounded when $|\tilde{\theta}|$ and ψ_e converges to a bounded residual set. The bound of the path following error is constrained in the compact as follows

$$\Omega_{y_e} = \left\{ y_e \mid |y_e| < \frac{\Delta M_e \sqrt{1 + \tan^2 \delta_\psi}}{\sigma} \right\} \quad \square \quad (60)$$

5. Simulation results and analysis

In this section, simulation experiments are carried out to evaluate the effectiveness of the path following system in this paper. The main parameters of the autonomous airship are available in [4].

5.1. The selection of the parameters in the simulation

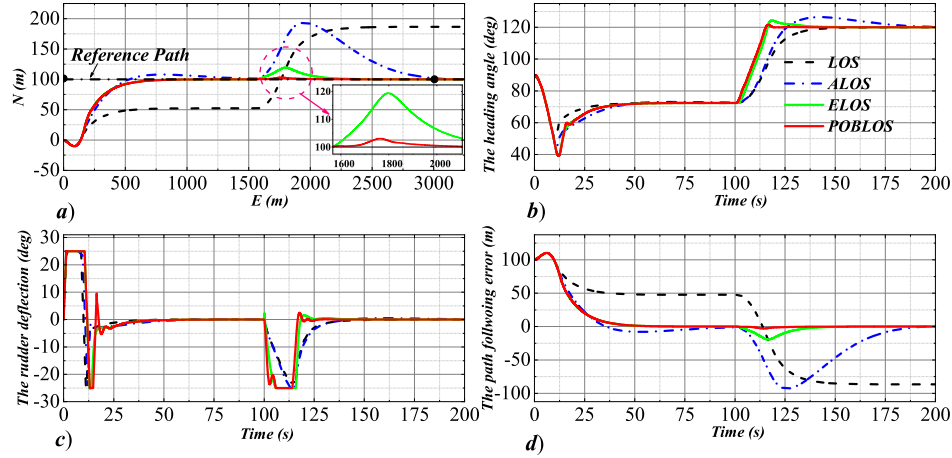
The parameter Δ is the so-called lookahead distance in LOS guidance law [11], which is selected according to the maneuvering characteristics of the airship. $\Delta = 150$ is adopted in this paper. The parameter $\delta_\psi = \pi/3$ is used to constrain the magnitude of the heading command. The selection of the control loop parameters (k_ψ , k_r and k_u) should satisfy the requirement of the stability analysis below (50). In order to realize a good dynamic performance, the controller gains are selected as $k_\psi = 1.2$, $k_r = 2.5$ and $k_u = 0.9$. The selection of the observer parameters (k_{y_e} , l_θ , k_{r_e} , l_r , k_{u_e} and l_u) should satisfy the stability conditions in (24) and (36). According to the conclusion below (21), the estimation error can be reduced by selecting the observer parameters appropriately under the constraints of the bandwidths and the dynamic performance [24]. In this way, the parameters of the disturbance observers are selected as $k_{r_e} = 10$, $l_r = 20$, $k_{u_e} = 6$ and $l_u = 10$. In order to avoid the aggressive actions, the bandwidth of the parameter observer should be matched with the slow dynamic characteristic of the airship in the guidance loop. As a result, the parameters of the parameter observer are selected as $k_{y_e} = 0.05$ and $l_\theta = 0.1$ to reduce the parameter estimation error efficiently in the meantime.

The related parameters of the command filters, the rudders and the propellers used in the simulation are arranged as Table 1.

Table 1

The parameters of the command filters, the rudder, and the propeller.

Model	Input variable	ω_n , rad/s	ξ	Magnitude limit	Rate limit
Command filter	ψ_c	20	0.7	$[\psi_p - 30, \psi_p + 30]$ deg	$[-5, 5]$ deg/s
	r_c	30	0.7	$[-5, 5]$ deg/s	–
	u_c	15	0.7	–	–
Rudder	δ_{rc}	40	0.75	$[-25, 25]$ deg	$[-20, 20]$ deg/s
Propeller	F_{Tc}	0.2	–	$[0, 5000]$ N	$[-1000, 1000]$ N/s

**Fig. 4.** The simulation comparisons of the different path following guidance laws: a) trajectory, b) heading angle, c) rudder deflection, d) path following error.

5.2. The effectiveness evaluation of the path following guidance law

In this subsection, simulation experiments are conducted to evaluate the effectiveness of POBLOS guidance law proposed in this paper. As comparisons, the traditional LOS guidance law, the adaptive integral LOS guidance law (ALOS) [11] and the reduced-order extended-state-observer based LOS guidance law (ELOS) [13] are adopted in the simulation. The design procedure and the related parameters of ALOS and ELOS are based on the design principles in [11] and [13], respectively. The control loop designed in this paper is used to execute the commands which are generated by the path following guidance laws mentioned above.

In this simulation case, the airship starts from the origin (0, 0) and moves towards the east at the initial speed of 15 m/s. The reference path is specified by the waypoints $w_{p1}(100, 0)$ and $w_{p2}(100, 3000)$. The desired velocity of the airship while following the path is set as 15 m/s. In the beginning, the wind is coming from the north and the wind speed is 5 m/s. After 100 s the wind is changed as

$$w_e = 0, \quad w_n = \begin{cases} -5, & t < 100 \\ -5 + (t - 100), & 100 \leq t < 115 \\ 10, & t \geq 115 \end{cases} \quad (61)$$

The simulation results are shown in Fig. 4 and Fig. 5.

In Fig. 4, the simulation results of the different path following guidance laws are summarized. Fig. 4a shows the trajectory of the airship. In the beginning, the trajectory is turning to the south due to the influence of the wind from the north. It can be seen that the wind has an influence on the shape of the trajectory, especially when the wind is time-varying from 100 s to 115 s. It is obvious that the path following error is sensitive to the wind when using the traditional LOS guidance law, while ALOS, ELOS and POBLOS can guarantee that the ability of convergence to the reference path, which is indicated in Fig. 4d. However, it takes a long time to converge to the reference path using ALOS and the results can

also be verified according to [11]. ELOS has a good performance, but the large deviation from the reference path occurs when the wind is time-varying, whereas POBLOS can still work well. Fig. 4b and Fig. 4c summarize the heading angle and the rudder deflection in the process of the path following, respectively. According to the results in Fig. 4b and Fig. 4d, ELOS and POBLOS can both guarantee that the heading angle is more sensitive to the path following error, which results in a better path following performance. It should be noted that the main design principles of ALOS, ELOS, and POBLOS are all trying to estimate the influence of the wind and compensate it to reduce the path following error. In this paper, the influence of wind is transformed into an unknown parameter θ which is described in (26). In order to reject the influence of the time-varying unknown parameter θ , the rudder deflection is more aggressive using ELOS and POBLOS according to the results in Fig. 4c and Fig. 5. When the unknown parameter θ is constant, the rudder deflection will converge to zero.

The change of the unknown parameter θ and its estimation value $\hat{\theta}$ are shown in Fig. 5. The effectiveness of the estimators/observers of the three guidance laws is compared. It is indicated that the convergence speed of estimation error is slow using the estimator of ALOS. The observer of ELOS has an excellent estimation of the unknown parameter θ when it is constant. However, the estimation error exists when the unknown parameter θ is time-varying. The unknown parameter θ can be approximated by $\hat{\theta}$ from the parameter observer of POBLOS even though θ is time-varying. It is inevitable that the small estimation error may exist when θ change abruptly, but the estimation error will converge gradually. The estimation information can be used to compensate for the influence of the wind, which results in a more robust performance of POBLOS.

5.3. The effectiveness evaluation of the control loop

In this simulation case, the effectiveness of the control loop is evaluated using three control methods: CFBC from [20], CFBC

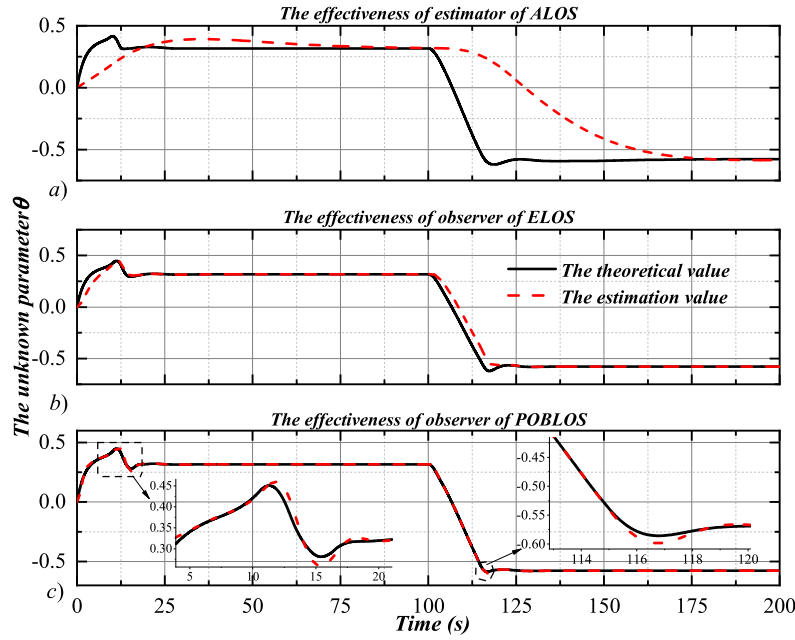


Fig. 5. The effectiveness of the estimators/observers: a) ALOS, b) ELOS, c) POBLOS.

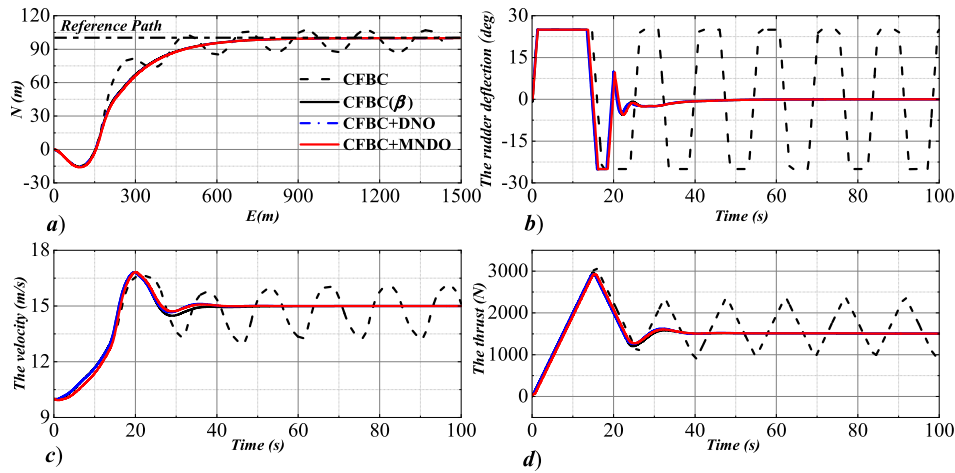


Fig. 6. The comparisons of different control methods when the wind speed is constant: a) trajectory, b) rudder deflection, c) velocity, d) thrust.

combined with NDO (CFBC + NDO) from [24] and CFBC based on MNDO (CFBC + MNDO) proposed in this paper. The initial velocity is 10 m/s and the other initial conditions are the same as those in section 5.2. The heading command is generated using POBLOS guidance law and the desired velocity is 15 m/s.

The wind is coming from the north and the wind speed is 5 m/s. The simulation results are summarized in Fig. 6.

In Fig. 6, CFBC(β) represents CFBC with the assumption that the sideslip angle β is known, which is used as a comparison to other methods. Fig. 6a describes the trajectory of the airship during the path following. It can be seen that CFBC has a poor performance in the presence of the wind because the backstepping controller is designed without knowing the information about the wind as described in Remark 1. The rudder deflection δ_r , the velocity u , and the thrust F_T during the path following are shown in Fig. 6b–6d, respectively. The influence of the wind results in the unknown sideslip angle β which has a serious impact on the dynamics of the airship. The problem will be solved if the sideslip angle β is known, which is indicated by the results of CFBC(β). However, it is usually difficult to obtain the sideslip angle β in the realistic

flight. By introducing the disturbance observer, we can see that the performance of CFBC is improved significantly. CFBC + NDO and CFBC + MNDO both work well and have a similar performance under the influence of the constant wind.

In order to verify the effectiveness of MNDO, the comparison of CFBC + NDO and CFBC + MNDO is carried out when the modeling uncertainties and the time-varying wind are taken into consideration. The aerodynamic coefficients are changed 50% larger than the original data. The mass and the inertia parameters in (2) are decreased 20% to represent the leak of the internal gas of the airship. The time-varying wind field is taken as (61) and the simulation results are depicted in Fig. 7.

In Fig. 7, CFBC + NDO(β) represents the method CFBC + NDO with the assumption that the sideslip angle β is known and is used as a comparison. Due to using the same guidance scheme, we can see that the trajectories of the airship are almost the same in Fig. 7a. According to Fig. 7c, the velocity of the airship can be stabilized to the desired velocity and the change of the thrust is given in Fig. 7d. The results of the rudder deflection during the path following are summarized in Fig. 7b. It can be seen

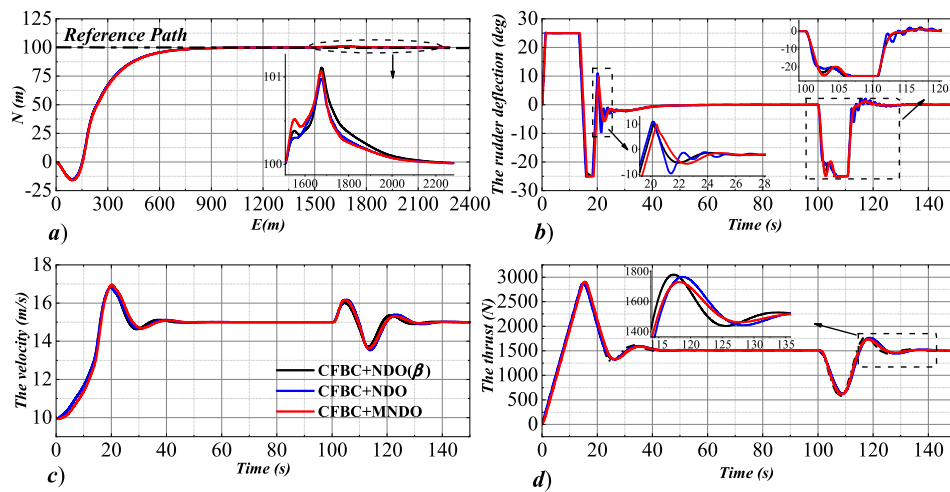


Fig. 7. The simulation results when uncertainties and time-varying wind are considered: a) trajectory, b) rudder deflection, c) velocity, d) thrust. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

that the oscillation of rudder deflection exists when CFBC + NDO is applied, whereas the oscillation will be reduced if the sideslip angle information is available, which is shown from the result of CFBC + NDO(β). It is indicated that the estimation ability of NDO is limited and the oscillation is induced by the estimation error of the sideslip angle. CFBC + MNDO can still work well even though under this severe circumstance and the performance is similar to CFBC + NDO(β), which indicates that CFBC + MNDO has a better performance compared to CFBC + NDO when uncertainties and the time-varying wind are taken into consideration.

6. Conclusion

The path following of the autonomous airship has been studied in this paper. The path following system is designed to keep the airship following the reference path in the presence of the wind and modeling uncertainties. In the guidance loop, the POB-LOS guidance law is proposed by combining the LOS guidance law with a parameter observer. The influence of the wind is expressed in terms of an unknown and time-varying parameter and estimated by the parameter observer. The control loop is designed based on the method of command filtered backstepping control combined with the disturbance compensation. The modeling uncertainties and time-varying disturbances are estimated by MNDO which is constructed based on the formulation of NDO and improved by adding the status estimation error feedback. The stability analyses of the guidance and control loops are conducted and the path following error is proved to be ultimately bounded. The simulation experiments are carried out to evaluate the effectiveness of the path following system. The simulation results indicate that the autonomous airship will converge to the reference path within the appropriate path following error.

Declaration of competing interest

There is no competing interest with any person or entity concerning the research reported in this paper.

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